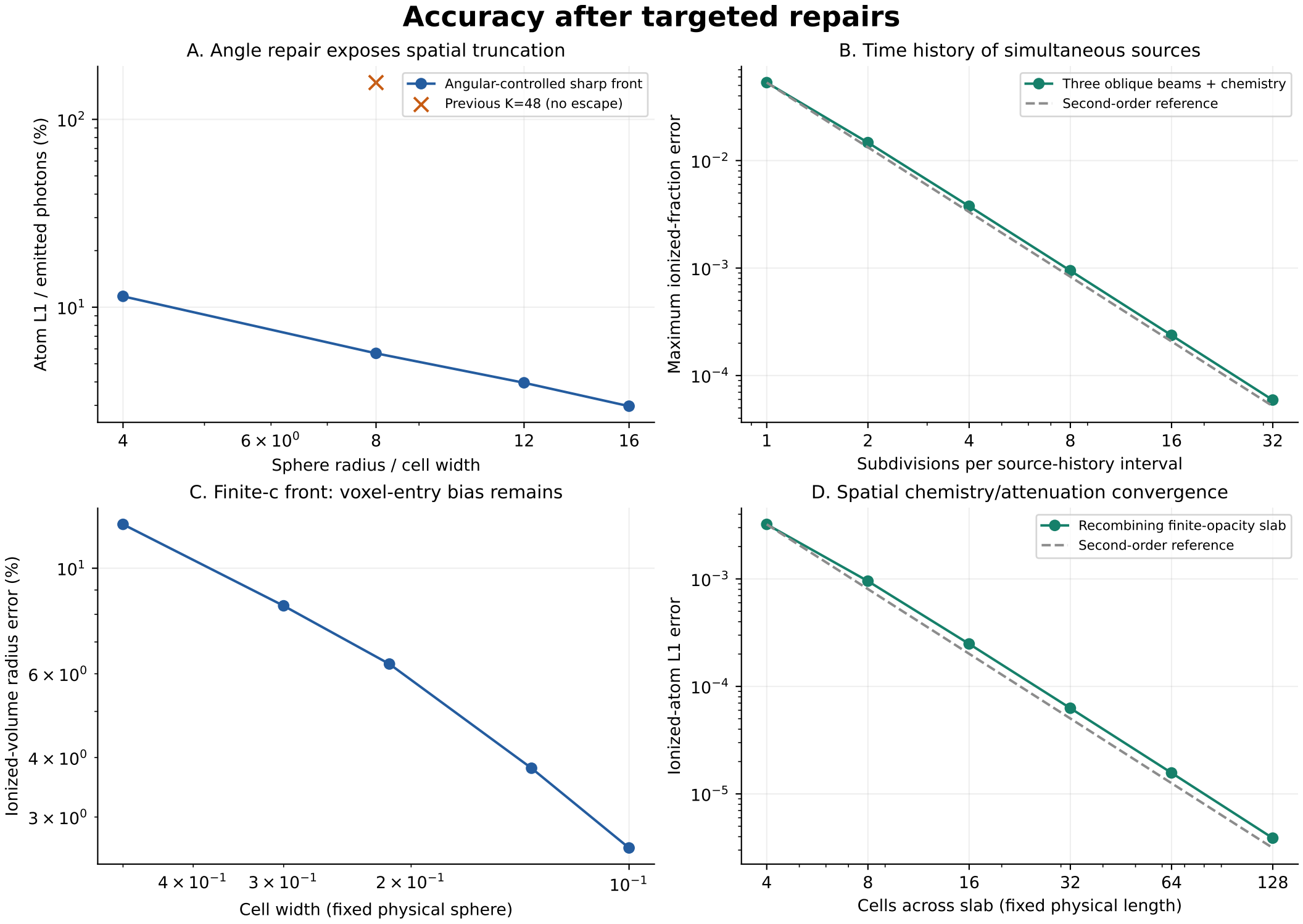
Accuracy after targeted repairs
A. Angle repair exposes spatial truncation
Angular-controlled sharp front
Previous K=48 (no escape)
Atom L1 / emitted photons (%)
10^2
10^1
4
6×10^0
8
12
16
Sphere radius / cell width
B. Time history of simultaneous sources
Three oblique beams + chemistry
Second-order reference
Maximum ionized-fraction error
10^{-2}
10^{-3}
10^{-4}
1
2
4
8
16
32
Subdivisions per source-history interval
C. Finite-c front: voxel-entry bias remains
Ionized-volume radius error (%)
10^1
6×10^0
4×10^0
3×10^0
4×10^{-1}
3×10^{-1}
2×10^{-1}
10^{-1}
Cell width (fixed physical sphere)
D. Spatial chemistry/attenuation convergence
Recombining finite-opacity slab
Second-order reference
Ionized-atom L1 error
10^{-3}
10^{-4}
10^{-5}
4
8
16
32
64
128
Cells across slab (fixed physical length)